S20822 PROGRAMMING ASSIGNMENT02 CSC4093

```
algal_leaf -> 113 images
brown_blight -> 113 images
white_spot -> 142 images
```

```
TRAIN SET
algal_leaf -> 79
brown_blight -> 79
white_spot -> 99

VAL SET
algal_leaf -> 16
brown_blight -> 16
white_spot -> 21

TEST SET
algal_leaf -> 18
brown_blight -> 18
white_spot -> 22
```

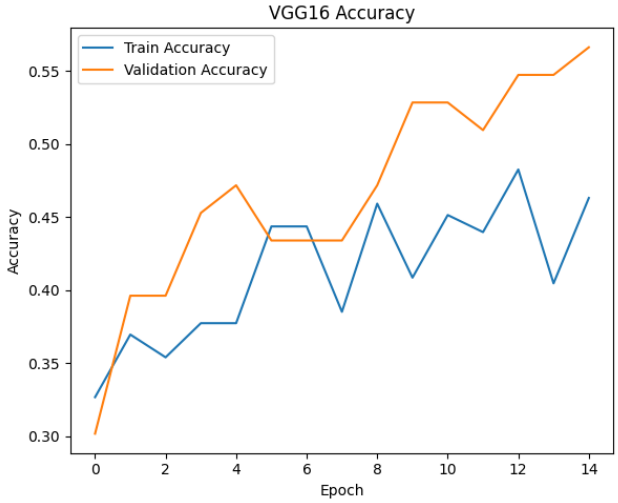
```
Found 257 images belonging to 3 classes.
Found 53 images belonging to 3 classes.
Found 58 images belonging to 3 classes.
```

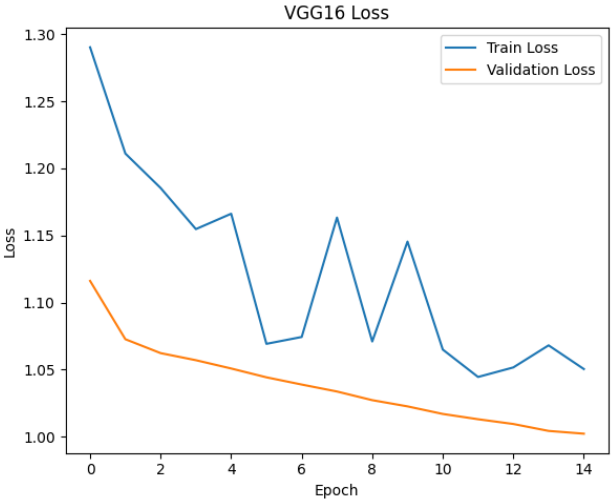
01.VGG16 Model:

```
Total params: 14,780,739 (56.38 MB)

Trainable params: 66,051 (258.01 KB)

Non-trainable params: 14,714,688 (56.13 MB)
```

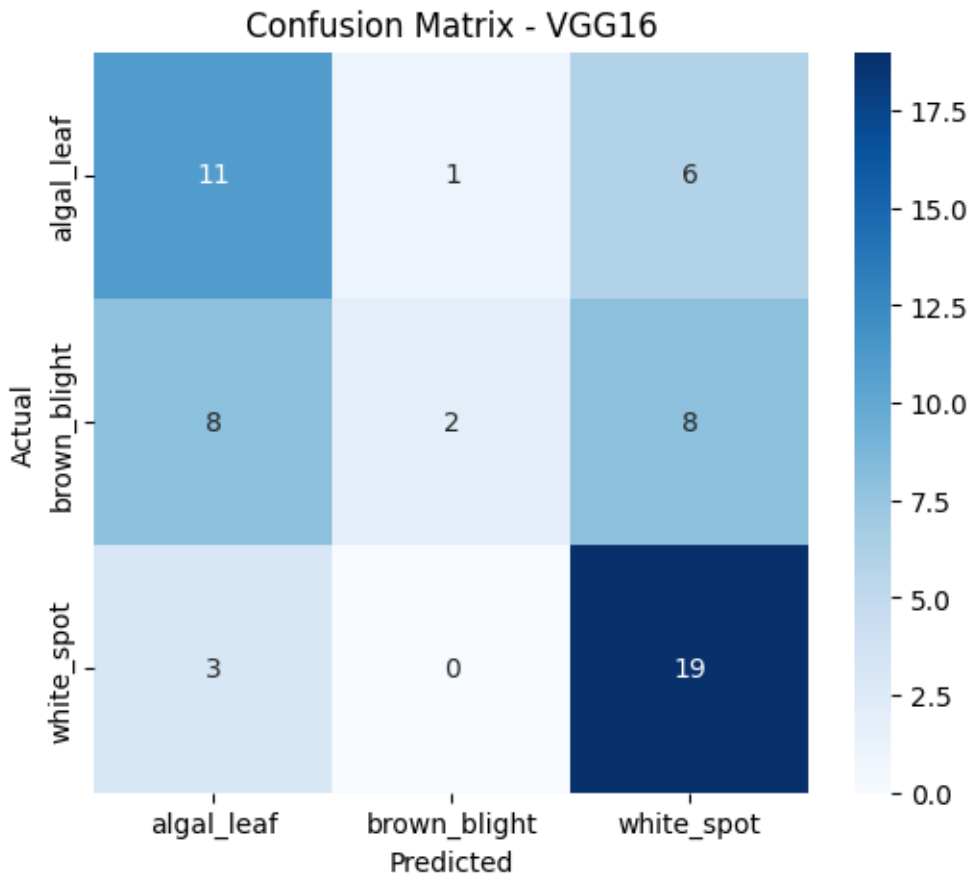
```
Epoch 1/15
17/17 ━━━━━━━━━━━━━━━━━━━━ 38s 2s/step - accuracy: 0.3268 - loss: 1.2902 - val_accuracy: 0.3019 - val_loss: 1.1161
Epoch 2/15
17/17 ━━━━━━━━━━━━━━━━━━━━ 35s 2s/step - accuracy: 0.3696 - loss: 1.2110 - val_accuracy: 0.3962 - val_loss: 1.0727
Epoch 3/15
17/17 ━━━━━━━━━━━━━━━━━━━━ 38s 2s/step - accuracy: 0.3541 - loss: 1.1853 - val_accuracy: 0.3962 - val_loss: 1.0623
Epoch 4/15
17/17 ━━━━━━━━━━━━━━━━━━━━ 35s 2s/step - accuracy: 0.3774 - loss: 1.1547 - val_accuracy: 0.4528 - val_loss: 1.0570
Epoch 5/15
17/17 ━━━━━━━━━━━━━━━━━━━━ 35s 2s/step - accuracy: 0.3774 - loss: 1.1662 - val_accuracy: 0.4717 - val_loss: 1.0509
Epoch 6/15
17/17 ━━━━━━━━━━━━━━━━━━━━ 34s 2s/step - accuracy: 0.4436 - loss: 1.0692 - val_accuracy: 0.4340 - val_loss: 1.0443
Epoch 7/15
17/17 ━━━━━━━━━━━━━━━━━━━━ 40s 2s/step - accuracy: 0.4436 - loss: 1.0744 - val_accuracy: 0.4340 - val_loss: 1.0389
Epoch 8/15
17/17 ━━━━━━━━━━━━━━━━━━━━ 56s 3s/step - accuracy: 0.3852 - loss: 1.1633 - val_accuracy: 0.4340 - val_loss: 1.0337
Epoch 9/15
17/17 ━━━━━━━━━━━━━━━━━━━━ 57s 4s/step - accuracy: 0.4591 - loss: 1.0710 - val_accuracy: 0.4717 - val_loss: 1.0273
Epoch 10/15
17/17 ━━━━━━━━━━━━━━━━━━━━ 56s 3s/step - accuracy: 0.4086 - loss: 1.1454 - val_accuracy: 0.5283 - val_loss: 1.0226
Epoch 11/15
17/17 ━━━━━━━━━━━━━━━━━━━━ 56s 3s/step - accuracy: 0.4514 - loss: 1.0650 - val_accuracy: 0.5283 - val_loss: 1.0170
Epoch 12/15
17/17 ━━━━━━━━━━━━━━━━━━━━ 56s 3s/step - accuracy: 0.4397 - loss: 1.0446 - val_accuracy: 0.5094 - val_loss: 1.0131
Epoch 13/15
...
Epoch 14/15
17/17 ━━━━━━━━━━━━━━━━━━━━ 57s 3s/step - accuracy: 0.4047 - loss: 1.0681 - val_accuracy: 0.5472 - val_loss: 1.0044
Epoch 15/15
17/17 ━━━━━━━━━━━━━━━━━━━━ 56s 3s/step - accuracy: 0.4630 - loss: 1.0505 - val_accuracy: 0.5660 - val_loss: 1.0023
```

Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...






```
4/4 ━━━━━━━━━━━━━━━━━━━━ 7s 2s/step - accuracy: 0.5517 - loss: 0.9993
VGG16 Test Loss: 0.9993290901184082
VGG16 Test Accuracy: 0.5517241358757019
```

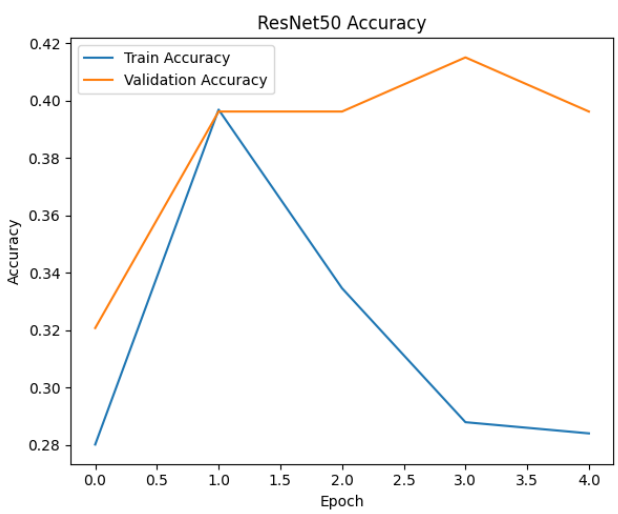


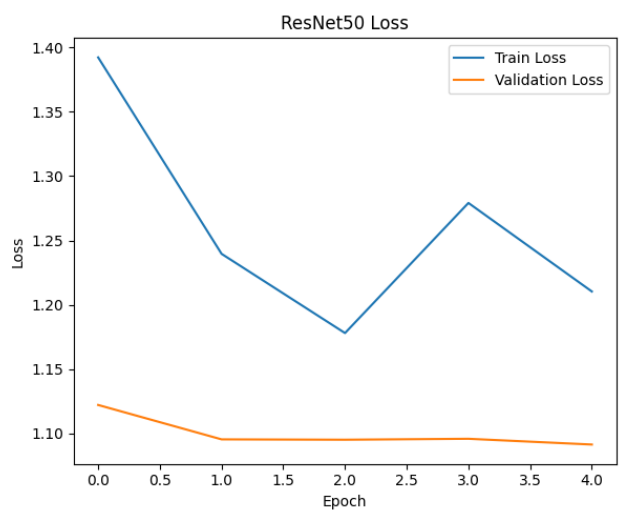
	precision	recall	f1-score	support
algal_leaf	0.50	0.61	0.55	18
brown_blight	0.67	0.11	0.19	18
white_spot	0.58	0.86	0.69	22
accuracy			0.55	58
macro avg	0.58	0.53	0.48	58
weighted avg	0.58	0.55	0.49	58

02.ResNet50 Model:

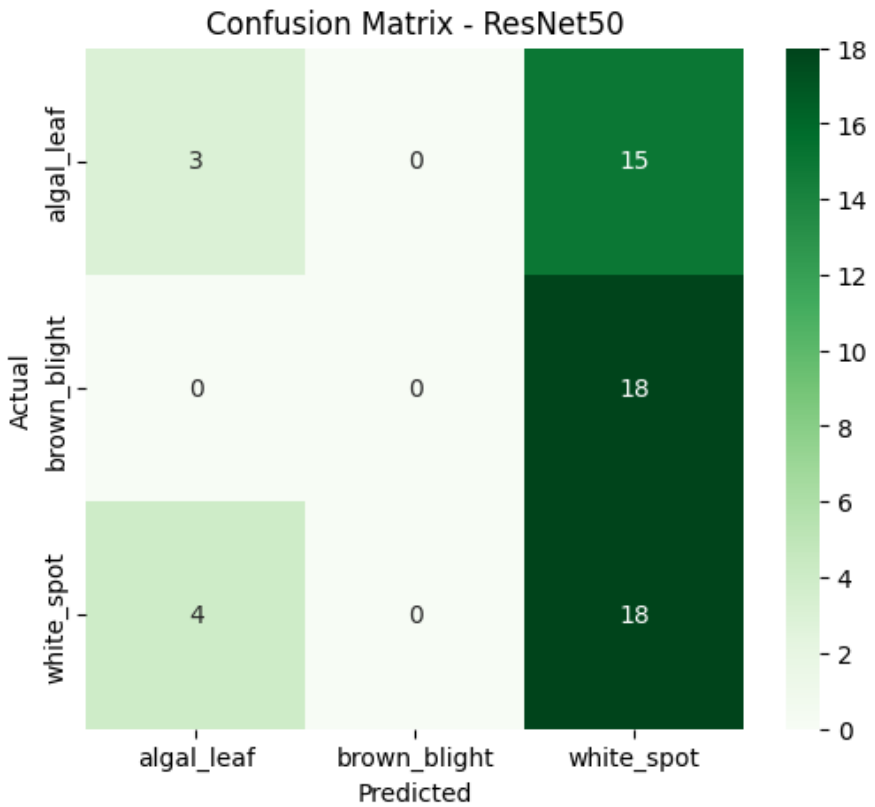
```
Epoch 1/15
17/17 ━━━━━━━━━━━━━━━━━━━━ 23s 1s/step - accuracy: 0.2802 - loss: 1.3923 - val_accuracy: 0.3208 - val_loss: 1.1221
Epoch 2/15
17/17 ━━━━━━━━━━━━━━━━━━━━ 14s 846ms/step - accuracy: 0.3969 - loss: 1.2396 - val_accuracy: 0.3962 - val_loss: 1.0954
Epoch 3/15
17/17 ━━━━━━━━━━━━━━━━━━━━ 14s 806ms/step - accuracy: 0.3346 - loss: 1.1780 - val_accuracy: 0.3962 - val_loss: 1.0951
Epoch 4/15
17/17 ━━━━━━━━━━━━━━━━━━━━ 14s 789ms/step - accuracy: 0.2879 - loss: 1.2792 - val_accuracy: 0.4151 - val_loss: 1.0959
Epoch 5/15
17/17 ━━━━━━━━━━━━━━━━━━━━ 13s 784ms/step - accuracy: 0.2840 - loss: 1.2103 - val_accuracy: 0.3962 - val_loss: 1.0914
```

```
Total params: 24,375,691 (92.99 MB)

Trainable params: 262,659 (1.00 MB)

Non-trainable params: 23,587,712 (89.98 MB)

Optimizer params: 525,320 (2.00 MB)
```





```
4/4 ━━━━━━━━━━━━━━━━━━━━ 3s 578ms/step - accuracy: 0.3621 - loss: 1.1220
ResNet50 Test Loss: 1.122023582458496
ResNet50 Test Accuracy: 0.36206895112991333
```




```
Classification Report - ResNet50

              precision    recall  f1-score   support

   algal_leaf       0.43      0.17      0.24        18
 brown_blight       0.00      0.00      0.00        18
   white_spot       0.35      0.82      0.49        22

     accuracy                           0.36        58
    macro avg       0.26      0.33      0.24        58
 weighted avg       0.27      0.36      0.26        58
```

Compare Both Model:

	Model	Test Accuracy	Test Loss
0	VGG16	0.551724	0.999329
1	ResNet50	0.362069	1.122024